

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Civil Engineering) SEMESTER - 6 Summer 2026 (Regular)

Course : Bachelor of Technology (Civil Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 6

Subject Code & Name: BTCVOE605J - PLANNING FOR SUSTAINABLE DEVELOPMENT

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No.6
4. Use of non-programmable scientific calculators is allowed.
5. Assume suitable data wherever necessary and mention it clearly.

Q1. Objective type questions. (Compulsory Question)

12

- 1 Sustainable development is best defined as:
 - A) Development without industrialization
 - B) Development that meets present needs without compromising future generations' needs
 - C) Development focused only on economic growth
 - D) Development without environmental protection
- 2 Which of the following is one of the three pillars of sustainable development?
 - A) Environmental protection
 - B) Military development
 - C) Space exploration
 - D) Industrial monopoly
- 3 The concept of sustainable development gained global recognition through the:
 - A) Paris Agreement
 - B) Kyoto Protocol
 - C) Brundtland Report
 - D) Montreal Protocol
- 4 Which of the following is a major cause of environmental degradation?
 - A) Afforestation
 - B) Rainwater harvesting
 - C) Recycling
 - D) Deforestation
- 5 Sustainable development seeks to:
 - A) Reduce poverty while protecting the environment
 - B) Increase pollution
 - C) Eliminate economic development
 - D) Promote resource depletion

- 6 Which international conference strongly promoted sustainable development?
 - A) Earth Summit (Rio de Janeiro, 1992)
 - ☒ B) Vienna Convention
 - C) Geneva Convention
 - D) Bretton Woods Conference
- 7 Renewable energy contributes to sustainable development by:
 - ☒ A) Depleting natural resources
 - B) Increasing greenhouse gas emissions
 - C) Reducing dependence on fossil fuels
 - D) Increasing pollution
- 8 Climate change is considered:
 - A) Unrelated to sustainability
 - ☒ B) A major issue affecting sustainable development
 - C) Only an economic issue
 - D) Only a political issue
- 9 Environmental management aims to:
 - A) Minimize environmental impacts while achieving development goals
 - B) Increase resource wastage
 - C) Promote environmental degradation
 - D) Reduce technological advancement
- 10 Sustainable innovation provides:
 - A) Environmental benefits only
 - ☒ B) Economic, environmental, and social benefits
 - C) Economic benefits only
 - D) Social benefits only
- 11 Capacity development refers to:
 - ☒ A) Strengthening skills, knowledge, and institutions for sustainable development
 - B) Increasing industrial pollution <https://www.batuonline.com>
 - C) Expanding urban areas only
 - D) Building larger factories only
- 12 Good governance for sustainable development is characterized by:
 - A) Transparency, accountability, and stakeholder participation
 - B) Secrecy and lack of accountability
 - C) Resource overexploitation
 - D) Environmental neglect

Q2. Solve the following.

- A) Define sustainable development. Explain its objectives, importance and the three pillars of sustainable development. 6
- B) Critically evaluate the concept of sustainable development. Discuss its relevance in the present global scenario. 6

Q3. Solve the following.

- A) Explain the relationship between environmental degradation and poverty. Discuss the major causes. 6
- B) Explain the main principles of sustainable development. Discuss the evolution of ideas related to sustainability. 6

Q4. Solve Any Two of the following.

- A) Explain various strategies for promoting sustainable development. 6
 - B) Discuss the major challenges and resistances to the implementation of sustainable development. 6
 - C) Explain the important current issues and areas of debate related to sustainable development. 6
- Q5. Solve Any Two of the following.
- A) Explain the role of innovation in achieving sustainable development. 6
 - B) Discuss environmental management strategies adopted for sustainable development. 6
 - C) Explain different innovation strategies that support environmental sustainability. 6
- Q6. Solve Any Two of the following.
- A) Explain the concept of societal transformation and its importance in sustainable development. 6
 - B) Discuss institutional theory and governance for sustainable development. 6
 - C) Explain policy responses to environmental degradation and discuss the importance of capacity development for innovation. 6

*** End ***